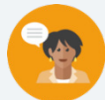


4.1 Understanding Set-BUILDER Notation

Lesson Introduction

 Benchmark	MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables, or written descriptions.		 Learning Targets	- I can understand the domain and range of a linear relationship. - I can write inequalities using set-builder notation.
 Benchmark Coherence	Building On MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.		Working Toward N/A	
 Essential Questions	- How can set-builder notation be used to describe the domain and range? - How are inequalities and set-builder notation the same? How are they different?			
 Lesson Narrative	This lesson activates prior knowledge regarding domain and range and positions students to make connections to a linear relationship using graphs and equations. Students will be introduced to set-builder notation for writing sets. The primary learning outcome in this lesson focuses on identifying and writing the domain and range of a linear relationship using inequalities and set-builder notation.			
 Component Summary	4.1.1 Warm-Up (~5 min.)	Students explore the concept of infinity (<i>SE p. 163</i>)	4.1.3 Guided Instruction (20 min.)	Introduction to set-builder notation (<i>SE p. 166</i>)
	4.1.2 Collaboration (15 min.)	Connect input/output values to domain/range vocabulary (<i>SE p. 164</i>)	4.1.4 Wrap-Up (~5 Min)	Students rate their understanding of domain, range, and set-builder notation (<i>SE p. 168</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Line - Domain - Range - Set-builder notation - Real numbers <u>Additional Glossary:</u> - Infinity		N/A	
<u>Additional Preparation</u> This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (Word Wall, Math Cards, etc.)				

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may get the domain and range backwards when applying the concept. - Students may misinterpret or misuse the inequality symbols when describing the domain and range. - Students may forget key symbols when writing in set-builder notation. - Students may forget to include infinity. - Students may overuse infinity in various situations.		- In questions 2-7, the equations that are referenced are the same as the ones graphed at the beginning of 4.1.2 Collaboration. - Set-builder notation will be used with regularity starting in Lesson 6; therefore, consider opportunities for spiraling practice throughout Lessons 1-5, rather than limited practice and proficiency to just this lesson.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Think-Pair-Share 4.1.2 Collaboration gives students the opportunity to work together. However, the questions are designed to spark thinking for the individual learner. Allow each student to make a conjecture about the answer to the questions and share their reasoning with a partner.		MA.K12.MTR.4.1: Discussion and Reflection 4.1.2 Collaboration gives students the opportunity to work together. Students get to work together to review pre-created fictional students' thinking to justify their thinking comparatively. They will analyze the mathematical thinking of others and justify results, gaining important practice in this Math Thinking and Reasoning Standard.	
 Differentiate Instruction	The graphs of the equations referenced during 4.1.2 Collaboration and 4.1.3 Guided Instruction at the beginning of each component can assist students who need a visual of the graph to assist with thinking through the questions. Auditory learners would benefit from hearing the verbal description of the written representations.			
Additional differentiation opportunities are denoted throughout the lesson guide.				
Lesson Notes:				

4.1.1 Warm-Up: What Is Infinity?

Component Overview: Students begin constructing meaningful entry points into mathematical discussions on the concept of infinity.



4.1.1 Warm-Up: What Is Infinity?

In *Toy Story*, Buzz Lightyear uses the phrase “To infinity and beyond.” But what does it mean to go beyond infinity? Infinity is an elusive concept that can often be difficult to comprehend. Eugenia Cheng, a mathematician, explains infinity as “not a normal kind of number.” Eugenia uses set theory to explain infinity. She uses the idea of a bag as a set. By placing a bag in a bag, there is a bag with one thing. Infinity can be illustrated by adding more and more bags and each bag contains other bags. Because you can keep adding bags with no end, the set contains an infinite number of bags.



1. Explain briefly your understanding of infinity.
Answers will vary. Infinity means to go on forever. Infinity means that it will never end.
2. In the video, Eugenia Cheng uses the word paradox. A paradox is a seemingly absurd or self-contradictory statement that when investigated may prove to be well-founded, or true. Why do you think the idea of infinity can create a paradox?
Answers will vary. Infinity can create a paradox because it is difficult to conceive an idea that something will never end. There are few things that never end that we experience daily, so the idea of infinity may seem absurd until investigated.
3. The symbol for infinity is ∞ . The infinity symbol was first used by John Wallis in 1655. Draw an infinity symbol without picking your writing utensil up from the paper.
Answers will vary. Drawings should accurately depict the infinity symbol. 
4. How many times can you trace over the infinity symbol without picking up your writing utensil from the paper?
Infinite; forever



At this point, students are sharing their understanding of the word “infinity”. This is not the place to expect the mathematical definition. 4.1.1 Warm-Up is working to move students toward making a connection between the general meaning of infinity and its use while determining the domain and range.

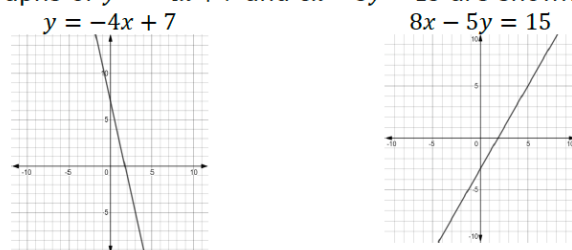
4.1.2 Collaboration: Domain and Range of Linear Relationships

Component Overview: Students will begin working with the domain and range of a linear equation (function language is not used in this lesson). The outcome of this section is for students to realize that there are situations where any value would be appropriate. Students will engage in a variety of MTR Standards in the collaborative activity. Consider giving students space and the opportunity for effective error analysis, math talk, and collaboration to orient students to the content and to each other.



4.1.2 Collaboration: Domain and Range of Linear Relationships

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are shown.



A **line** is a straight path that extends infinitely in both directions.

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are lines.

- Two students were discussing how to determine if there is an x -value(s) not included in the graph of $y = -4x + 7$. Their explanations are shown.

Quanasia	Lysha
The graph of $y = -4x + 7$ is steep so it will never reach values such as $x = 10$. I think it eventually reaches $x = 5$, but I don't think it reaches many x -values after that.	The graph cannot show the entire line. If I input any value in for x , the output, y , will always be a number.

Discuss with your partner the two different explanations. Then, use the definition of a line to justify whether there is an x -value(s) not included in the graph of the line $y = -4x + 7$.

Answers will vary. If a line extends infinitely in both directions, it would have an x - and a y -value. For a line to continue, it would need points and each point is in (x, y) form.



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.



What does it mean for something to extend infinitely in both directions?



What are the key differences in their arguments?



Students are not expected to have a perfect mathematical answer at this point. This concept is being developed, so allow students to conjecture.

Students have not had experience with the appropriate vocabulary to communicate this mathematical idea effectively yet (MA.K12.MTR.4.1). Since this concept is still being developed, allow students the opportunity to make a reasonable conjecture.

4.1.2 Collaboration: Domain and Range of Linear Relationships (continued)

2. Based on the definition of domain, what values can be used as an input for x if $y = -4x + 7$?
 - Ⓐ all values greater than -5 and less than 5
 - Ⓑ all values greater than -20 and less than 20
 - Ⓒ it cannot be determined
 - Ⓓ **any value**
3. Based on the definition of domain, what values can be used as an input for x if $8x - 5y = 15$?
 - Ⓐ all values greater than -5 and less than 5
 - Ⓑ all values greater than -20 and less than 20
 - Ⓒ it cannot be determined
 - Ⓓ **any value**
4. List any y -value(s) not included in the graph of $y = -4x + 7$. If none, then explain why.
Answers will vary. None because for any value I input for x , I will get a y -value. None because for every input, I will have an output.
5. List any y -value(s) not included in the graph of $8x - 5y = 15$. If none, then explain why.
Answers will vary. None because for any value I input for x , I will get a y -value. None because for every input, I will have an output.



The **range** is the complete set of possible values of the output of a relation or function.

6. Based on the definition of range, for the linear relationship $y = -4x + 7$, what values are outputs for y ?
 - Ⓐ all values greater than -10 and less than 15
 - Ⓑ all values greater than -20 and less than 20
 - Ⓒ it cannot be determined
 - Ⓓ **any value**
7. Based on the definition of range, for the linear relationship $8x - 5y = 15$, what values are outputs for y ?
 - Ⓐ all values greater than -10 and less than 10
 - Ⓑ all values greater than -20 and less than 20
 - Ⓒ it cannot be determined
 - Ⓓ **any value**



For a kinesthetic approach for collaboration for this activity, label each corner of the room as A, B, C, and D, then have students choose a corner based on their answer.

Each corner then chooses a spokesperson for their corner to provide rationale for their response.

Students can then choose to stay in their corner or re-shuffle based on their arguments.



Students may believe only what is pictured in the coordinate plane can be a value for the domain and range. In other words, the line does not extend beyond the plane they see. Consider choosing a value not pictured and proving its range using the equation.

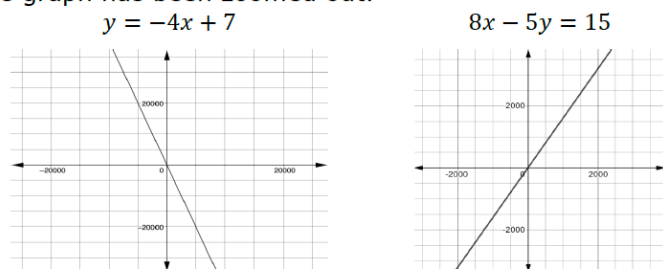
4.1.3 Guided Instruction: Set-Builder Notation

Component Overview: Students are introduced to the concept of set-builder notation. They will begin with an understanding of all real numbers as it relates to domain and range and move into representing that in written form, as an inequality, or in set-builder notation. Begin this activity by providing students an opportunity to make observations, prompting them to make connections between the two representations.



4.1.3 Guided Instruction: Set-Builder Notation

The graphs of $y = -4x + 7$ and $8x - 5y = 15$ are shown where the graph has been zoomed out.



The domain and range of a linear relationship are all **real numbers**.

By zooming out, it is clearer that the domain and range of $y = -4x + 7$ and $8x - 5y = 15$ is all real numbers.

There are several ways to mathematically represent the domain and range of all real numbers.

	Written Description	Inequalities	Set-Builder Notation
Domain	All real numbers	$-\infty < x < \infty$	$\{x: x \in R\}$ or $\{x x \in R\}$
Range	All real numbers	$-\infty < y < \infty$	$\{y: y \in R\}$ or $\{y y \in R\}$

The representation of the domain using inequalities is read as, "negative infinity is less than x , which is less than infinity."

Notice that the inequality sign used with infinity does not include an equal sign.



Students may notice the different scales and that the lines continue beyond the x - and y -values in the original graphs.



What do you notice about each line? (it keeps going) How does the term "all real numbers" relate to what we know about the word "infinity"? (all real numbers can be the mathematical representation of infinity in the context of domain and range)



Do not spend too much time on this table. The purpose is to show what the representations look like. As the lesson unfolds further, it will guide students through how they are represented, read, and written.

4.1.3 Guided Instruction: Set-Builder Notation (continued)

- Discuss with your partner why the domain $-\infty < x < \infty$ cannot be written as $-\infty \leq x \leq \infty$. *Answers will vary. If infinity goes on forever, it would not stop going, meaning nothing can be equal to something always continuing. Any number I can think of as a value, I can go one larger or smaller.*



Set-builder notation is a shorthand used to write sets, often sets with an infinite number of elements. The set $\{x: x > 0\}$ is read aloud, "the set of all x , such that x is greater than 0." It is read aloud exactly the same way when the colon (:) is replaced by the vertical line (|) as in $\{x|x > 0\}$.

The table shows number sets within the number system, examples of the types of numbers included in each set, and the symbol used in set-builder notation, if there is one.

	Examples	Symbol
Natural numbers	1, 2, 3, 4, ...	N
Whole numbers	0, 1, 2, 3, ...	No symbol
Integers	..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...	Z
Rational numbers	$-20\frac{7}{8}, -\frac{2}{7}, -0.25, \sqrt{9}, 4.01, \sqrt[3]{125}, 14\frac{5}{9}, 89$	Q
Irrational numbers	$-\sqrt{17}, \pi, \frac{\sqrt{6}}{2}, \sqrt[3]{28}$	No symbol
Real numbers	All rational and irrational numbers	R

In most cases, set-builder notation is similar to an inequality, except when referring to a number system or a specific pattern of numbers.



Allow students to engage in meaningful dialogue related to the symbol changes in this question so they can distinguish between them.



Why do you think mathematicians came up with specific symbols?

Why do you think whole numbers and irrational numbers have no symbol?

4.1.3 Guided Instruction: Set-Builder Notation (continued)

2. Work with your partner to complete the table. The first row has been completed.

Written Description	Inequalities	Set-Builder Notation
All numbers greater than or equal to negative three	$x \geq -3$	$\{x: x \geq -3\}$ or $\{x x \geq -3\}$
All numbers less than seven	$x < 7$	$\{x: x < 7\}$ or $\{x x < 7\}$
All numbers greater than or equal to negative two and less than or equal to eleven	$x \geq -2$ and $x \leq 11$ or $-2 \leq x \leq 11$	$\{x: -2 \leq x \leq 11\}$
All numbers greater than zero and less than or equal to twenty	$x > 0$ and $x \leq 20$ or $0 < x \leq 20$	$\{x 0 < x \leq 20\}$
The set of integers	n/a	$\{x x \in \mathbb{Z}\}$
The set of rational numbers	n/a	$\{x x \in \mathbb{Q}\}$

3. Explain why writing the set of integers and the set of rational numbers using inequalities is not possible.
Answers will vary. Using the infinity symbol would include all values, not just rational or integers. It would require an infinite number of specific inequalities without using the infinity symbol to be limited to one specific type of number.



This problem provides an opportunity to determine a student's understanding of turning a domain/range from a written description into inequalities and set-builder notation.

Students who struggle with the inequality form will likely struggle with set-builder notation.

Set-builder notation is new for all students, making it reasonable for most students to have errors in this section.



Provide an opportunity for meaningful reflection with question 3. This response could be coupled with 4.1.4 Wrap-Up for you to evaluate student progress from the lesson.

4.1.4 Wrap-Up: Reflection

Component Overview: Students complete a self-reflection on their level of understanding of the content in this lesson.



4.1.4 Wrap-Up: Reflection

1. Use the emoji scale to describe how well you understand the domain and range of a linear relationship.



0 1 2 3 4 5 6 7 8 9 10

Answers will vary.

2. Use the emoji scale to describe how well you understand set-builder notation.



0 1 2 3 4 5 6 7 8 9 10

Answers will vary.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



What would reflection look like for you as a teacher? Consider your own scale for strengths and gaps for students through the lesson and how best to spiral set-builder notation throughout Lessons 2-5 before this notation is used with regularity starting in Lesson 6.